Assignment 12

Problem Definition: Library Management System

Develop a simple Library Management System in Java. The system should maintain information about books and borrowers. A consumer should be able to search for books, borrow an available book, and return a borrowed book.

Requirements

- Develop a simple Library Management System in Java.
- Maintain information about books and borrowers.
- Allow consumers to search for books, borrow an available book, and return a borrowed book.
- Create a Book class with private title, author, availability, and borrower attributes.
- Implement setData, borrowBook, returnBook, and displayDetails methods.
- Create a Borrower class with private borrowerId and borrowerName attributes.
- Implement setter, getter, and display methods for Borrower.
- Create arrays of Book and Borrower objects in main().
- Initialize books and borrowers using setter methods.
- Display all books and borrowers.
- Search books by title and author.
- Store the Borrower object in the corresponding Book object when borrowed.
- Prevent borrowing a book that is already borrowed.
- Allow a borrower to return a book and remove the associated borrower information.
- Display the updated book details.

Suggested Menu

```
===== LIBRARY MANAGEMENT SYSTEM =====
1. Display All Books
2. Display All Borrowers
3. Search Book by Title
4. Search Book by Author
5. Borrow Book
6. Return Book
7. Exit
Enter your choice:
```

Source Code

```
import java.util.Scanner;

class Book {
    private String title;
    private String author;
    private Boolean isAvailable = true;
    private Borrower borrower;

    void setData(String title, String author) {
        this.title = title;
        this.author = author;
    }

    String getTitle() {
        return title;
    }

    String getAuthor() {
        return author;
    }

    void borrowBook(Borrower borrower) {
        if (isAvailable) {
            this.borrower = borrower;
            isAvailable = false;
        } else
            System.out.println("Book already borrowed by: " + this.borrower.getBorrowerName() + "\n");
    }

    void returnBook() {
        if (isAvailable) {
            System.out.println("This book is not borrowed yet");
            return;
        }
        this.borrower = null;
        isAvailable = true;
        System.out.println("Book successfully returned\n");
    }

    void displayDetails() {
        System.out.println("\nBook title: " + title);
        System.out.println("Book author: " + author);
        System.out.println("Availabilty: " + isAvailable);
        if (!isAvailable) {
            borrower.displayBorrowerDetails();
        }
    }
}

class Borrower {
    private int borrowerId;
    private String borrowerName;

    void setBorrowerId(int borrowerId) {
        this.borrowerId = borrowerId;
    }

    void setBorrowerName(String borrowerName) {
        this.borrowerName = borrowerName;
    }

    int getBorrowerId() {
        return borrowerId;
    }

    String getBorrowerName() {
        return borrowerName;
    }
```

```
    void displayBorrowerDetails() {
        System.out.println("Borrower Name: " + borrowerName);
        System.out.println("Borrower ID: " + borrowerId);
    }
}

public class Main {
    public static void main(String[] args) {
        Book[] books = new Book[4];
        Borrower[] borrowers = new Borrower[4];

        for (int i = 0; i < 4; i++) {
            books[i] = new Book();
            borrowers[i] = new Borrower();
            borrowers[i].setBorrowerId(i + 1);
        }

        books[0].setData("Attack on Titan", "Hajime Isayama");
        books[1].setData("Death Note", "Tsugumi Ohba");
        books[2].setData("Naruto", "Masashi Kishimoto");
        books[3].setData("Akira", "Katsuhiro Otomo");

        borrowers[0].setBorrowerName("Zaraki Kenpachi");
        borrowers[1].setBorrowerName("Kuchiki Rukia");
        borrowers[2].setBorrowerName("Yami Sukehiro");
        borrowers[3].setBorrowerName("William Vanjance");

        boolean run = true;
        Scanner sc = new Scanner(System.in);
        while (run) {
            System.out.println("\n===== LIBRARY MANAGEMENT SYSTEM =====");
            System.out.println("\n1. Display All Books");
            System.out.println("2. Display All Borrowers");
            System.out.println("3. Search Book by Title");
            System.out.println("4. Search Book by Author");
            System.out.println("5. Borrow Book");
            System.out.println("6. Return Book");
            System.out.println("7. Exit\n");
            System.out.print("Enter your choice: ");
            int choice = Integer.parseInt(sc.nextLine());
            int i, j;
            switch (choice) {
                case 1:
                    for (i = 0; i < 4; i++) {
                        books[i].displayDetails();
                    }
                    break;

                case 2:
                    for (i = 0; i < 4; i++) {
                        borrowers[i].displayBorrowerDetails();
                    }
                    break;

                case 3:
                    System.out.println("Enter book title: ");
                    String bookName = sc.nextLine();
                    for (i = 0; i < books.length; i++) {
                        if (books[i].getTitle().toLowerCase().contains(bookName.strip().toLowerCase())) {
                            books[i].displayDetails();
                            break;
                        }
                    }
                    if (i == books.length)
                        System.out.println("Book not present");
                    break;

                case 4:
                    System.out.println("Enter book author: ");
                    String authorName = sc.nextLine();
                    for (i = 0; i < books.length; i++) {
```

```
                    if (books[i].getAuthor().toLowerCase().contains(authorName.strip().toLowerCase())) {
                        books[i].displayDetails();
                        break;
                    }
                }
                if (i == books.length)
                    System.out.println("Book not present");
                break;

            case 5:
                System.out.println("Enter book name: ");
                String book = sc.nextLine();
                for (i = 0; i < books.length; i++) {
                    if (books[i].getTitle().toLowerCase().contains(book.strip().toLowerCase())) {
                        System.out.println("Enter borrower's name: ");
                        String borrowerName = sc.nextLine();
                        for (j = 0; j < borrowers.length; j++) {
                            if (borrowers[j].getBorrowerName().toLowerCase()
                                    .contains(borrowerName.strip().toLowerCase())) {
                                books[i].borrowBook(borrowers[j]);
                                break;
                            }
                        }
                        break;
                    }
                }
                break;

            case 6:
                System.out.println("Enter book name: ");
                String Book = sc.nextLine();
                for (i = 0; i < books.length; i++) {
                    if (books[i].getTitle().toLowerCase().contains(Book.strip().toLowerCase())) {
                        books[i].returnBook();
                        break;
                    }
                }
                break;

            case 7:
                run = false;
                break;

            default:
                break;
            }
        }
        sc.close();
    }
}
```

Output

```
===== LIBRARY MANAGEMENT SYSTEM =====

1. Display All Books
2. Display All Borrowers
3. Search Book by Title
4. Search Book by Author
5. Borrow Book
6. Return Book
7. Exit

Enter your choice: 1

Book title: Attack on Titan
Book author: Hajime Isayama
Availabilty: true

Book title: Death Note
Book author: Tsugumi Ohba
Availabilty: true

Book title: Naruto
Book author: Masashi Kishimoto
Availabilty: true

Book title: Akira
Book author: Katsuhiro Otomo
Availabilty: true

===== LIBRARY MANAGEMENT SYSTEM =====

1. Display All Books
2. Display All Borrowers
3. Search Book by Title
4. Search Book by Author
5. Borrow Book
6. Return Book
7. Exit

Enter your choice: 2
Borrower Name: Zaraki Kenpachi
Borrower ID: 1
Borrower Name: Kuchiki Rukia
Borrower ID: 2
Borrower Name: Yami Sukehiro
Borrower ID: 3
Borrower Name: William Vanjance
Borrower ID: 4

===== LIBRARY MANAGEMENT SYSTEM =====

1. Display All Books
2. Display All Borrowers
3. Search Book by Title
4. Search Book by Author
5. Borrow Book
6. Return Book
7. Exit

Enter your choice: 3
Enter book title:
Naruto

Book title: Naruto
Book author: Masashi Kishimoto
Availabilty: true
```

```
===== LIBRARY MANAGEMENT SYSTEM =====

1. Display All Books
2. Display All Borrowers
3. Search Book by Title
4. Search Book by Author
5. Borrow Book
6. Return Book
7. Exit

Enter your choice: 4
Enter book author:
Kishimoto

Book title: Naruto
Book author: Masashi Kishimoto
Availabilty: true

===== LIBRARY MANAGEMENT SYSTEM =====

1. Display All Books
2. Display All Borrowers
3. Search Book by Title
4. Search Book by Author
5. Borrow Book
6. Return Book
7. Exit

Enter your choice: 5
Enter book name:
Death Note
Enter borrower's name:
Yami Sukehiro

===== LIBRARY MANAGEMENT SYSTEM =====

1. Display All Books
2. Display All Borrowers
3. Search Book by Title
4. Search Book by Author
5. Borrow Book
6. Return Book
7. Exit

Enter your choice: 1

Book title: Attack on Titan
Book author: Hajime Isayama
Availabilty: true

Book title: Death Note
Book author: Tsugumi Ohba
Availabilty: false
Borrower Name: Yami Sukehiro
Borrower ID: 3

Book title: Naruto
Book author: Masashi Kishimoto
Availabilty: true

Book title: Akira
Book author: Katsuhiro Otomo
Availabilty: true

===== LIBRARY MANAGEMENT SYSTEM =====
```

```
1. Display All Books
2. Display All Borrowers
3. Search Book by Title
4. Search Book by Author
5. Borrow Book
6. Return Book
7. Exit

Enter your choice: 6
Enter book name:
Naruto
This book is not borrowed yet

===== LIBRARY MANAGEMENT SYSTEM =====

1. Display All Books
2. Display All Borrowers
3. Search Book by Title
4. Search Book by Author
5. Borrow Book
6. Return Book
7. Exit

Enter your choice: 5
Enter book name:
Death Note
Enter borrower's name:
William
Book already borrowed by: Yami Sukehiro

===== LIBRARY MANAGEMENT SYSTEM =====

1. Display All Books
2. Display All Borrowers
3. Search Book by Title
4. Search Book by Author
5. Borrow Book
6. Return Book
7. Exit

Enter your choice: 6
Enter book name:
Death Note
Book successfully returned

===== LIBRARY MANAGEMENT SYSTEM =====

1. Display All Books
2. Display All Borrowers
3. Search Book by Title
4. Search Book by Author
5. Borrow Book
6. Return Book
7. Exit

Enter your choice: 7
```